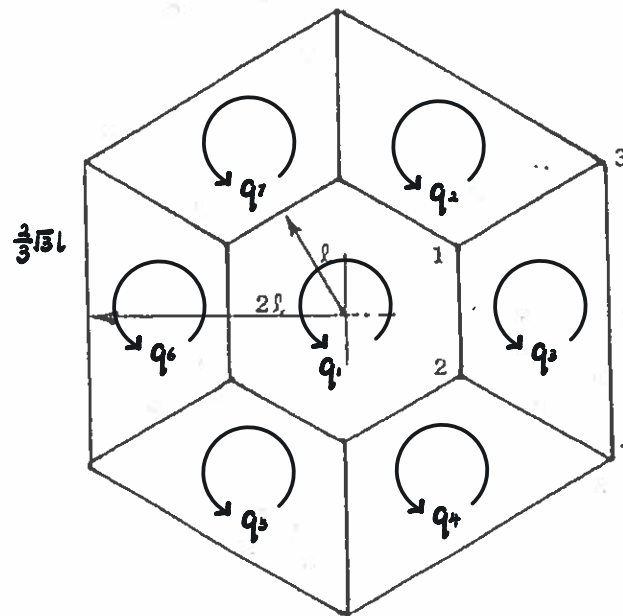


Tutorial Sheet 2 – Multicell tubes

1. The figure shows the cross-section of a thin-walled tube which is based upon a regular hexagon. The wall thickness is constant throughout.

Show that if the section is modified by removing walls 1-2 and 3-4 the torsion constant J is reduced to about $\frac{1}{4}$ of that of the unmodified section.



Before modifying.

according to the symmetry, $q_2 = q_3 = q_4 = q_5 = q_6 = q_1$

$$\begin{aligned} \therefore T &= 2 \sum_{i=1}^6 q_i A_i = 2 (6 q_2) \left(\frac{\frac{4}{3}\sqrt{3}l + \frac{2}{3}\sqrt{3}l}{2} \cdot l \right) + 2 q_1 \left(6 \cdot \frac{1}{2} \cdot \frac{2}{3}\sqrt{3}l \cdot l \right) \\ &= 12\sqrt{3}l^2 q_2 + 4\sqrt{3}l^2 q_1 \end{aligned}$$

$$\left(\frac{d\theta}{dz} \right)_2 = \frac{1}{2 \cdot \sqrt{3}l^2 \cdot G} \cdot \frac{1}{t} [(q_2 - q_1) \cdot \frac{2}{3}\sqrt{3}l + q_2 \cdot \frac{4}{3}\sqrt{3}l] = \frac{1}{Glt} (q_2 - \frac{1}{3}q_1)$$

$$\left(\frac{d\theta}{dz} \right)_1 = \frac{1}{2 \cdot \sqrt{3}l^2 \cdot G} \cdot \frac{1}{t} [6 (q_1 - q_2) \cdot \frac{2}{3}\sqrt{3}l] = \frac{1}{Glt} (q_1 - q_2)$$

$$\therefore q_2 - \frac{1}{3}q_1 = q_1 - q_2 \quad \therefore 2q_2 = \frac{4}{3}q_1 \quad \therefore q_2 = \frac{2}{3}q_1$$

$$\therefore T = 12\sqrt{3}l^2 q_1 \quad \therefore q_1 = \frac{T}{12\sqrt{3}l^2}$$

$$\therefore \frac{d\theta}{dz} = \frac{1}{Gl^3} \cdot \frac{1}{3} \frac{T}{12\sqrt{3}l^2} = \frac{T}{36\sqrt{3}Gl^3t}$$

$$\therefore J = \frac{T}{G\theta'} = \frac{T}{G} \cdot \frac{36\sqrt{3}Gl^3t}{T} = 36\sqrt{3}l^3t$$

After modifying,

according to the symmetry, $q_2 = q_4$, $q_7 = q_5$

$$\therefore T = 2(2q_2 + 2q_5 + q_6)(\sqrt{3}l^2)$$

$$\left(\frac{d\theta}{dz}\right)_2 = \frac{1}{2\sqrt{3}l^2 \cdot G} \frac{1}{t} \left[\left(\frac{2}{3}\sqrt{3}l + \frac{2}{3}\sqrt{3}l + \frac{4}{3}\sqrt{3}l\right) \cdot q_2 + (q_2 - q_1) \cdot \frac{2}{3}\sqrt{3}l \right]$$

$$= \frac{1}{Gl^3t} \left(\frac{5}{3}q_2 - \frac{1}{3}q_1 \right)$$

$$\left(\frac{d\theta}{dz}\right)_5 = \frac{1}{Gl^3t} \left(\frac{5}{3}q_5 - \frac{1}{3}q_2 - \frac{1}{3}q_6 \right)$$

$$\left(\frac{d\theta}{dz}\right)_6 = \frac{1}{Gl^3t} \left(\frac{5}{3}q_6 - \frac{2}{3}q_5 \right)$$

$$\therefore 5q_2 - q_5 = 5q_5 - q_2 - q_6 = 5q_6 - 2q_5$$

$$\therefore q_2 = q_5 - \frac{1}{6}q_6$$

$$\therefore 4q_5 - \frac{5}{6}q_6 = 5q_6 - 2q_5$$

$$\therefore q_5 = \frac{35}{36}q_6 \quad \therefore q_2 = \frac{29}{36}q_6$$

$$\therefore T = \frac{82}{9}\sqrt{3}l^2q_6 \quad \therefore q_6 = \frac{9T}{82\sqrt{3}l^2}$$

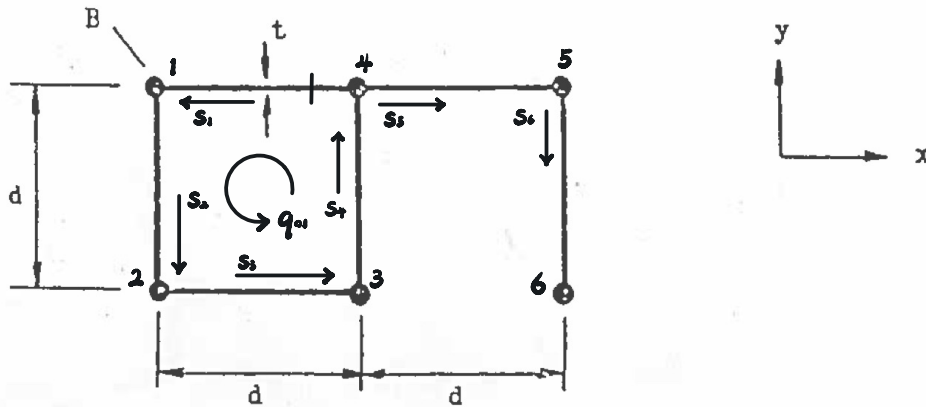
$$\therefore \frac{d\theta}{dz} = \frac{1}{Gl^3t} \left(\frac{55}{54}q_6 \right) = \frac{55T}{492\sqrt{3}Gl^3t}$$

$$\therefore J = \frac{T}{G\theta'} = \frac{T}{G} \cdot \frac{492\sqrt{3}Gl^3t}{55T} = \frac{492\sqrt{3}}{55}l^3t$$

$$\therefore \frac{J_m}{J_{um}} = \frac{41}{165} \approx \frac{1}{4}$$

2. A uniform tube has a cross-section composed of a closed cell and an open cell, as shown in the Figure below. The shape of the cross-section is maintained by the usual system of closely spaced ribs. The six booms, of area B , are assumed to carry only direct stresses and the sheet, of thickness t , is assumed to carry only shear stress.

Determine the position of the shear centre of the cross-section.



$$I_x = 3 \cdot \left(\frac{d}{2} - 0\right)^2 \cdot B + 3 \cdot \left(0 - \frac{d}{2}\right)^2 \cdot B = \frac{3}{2} B d^2$$

$$I_y = 2 \cdot (0 - d)^2 \cdot B + 2 \cdot 0 \cdot B + 2 \cdot (d - 0)^2 \cdot B = 4 B d^2$$

'cut' close-tube between 1-4 as shown above

$$(D_x)_{s1} = - \int_0^s \hat{y} t \, ds_1 = 0$$

$$(D_y)_{s1} = - \int_0^s \hat{x} t \, ds_1 = 0$$

$$(D_x)_{s2} = - \frac{1}{2} B d$$

$$(D_y)_{s2} = B d$$

$$(D_x)_{s3} = - \frac{1}{2} B d - (- \frac{1}{2} B d) = 0$$

$$(D_y)_{s3} = B d - (- B d) = 2 B d$$

$$(D_x)_{s4} = 0 - (- \frac{1}{2} B d) = \frac{1}{2} B d$$

$$(D_y)_{s4} = 2 B d$$

$$(D_x)_{s5} = \frac{1}{2} B d - (\frac{1}{2} B d) = 0$$

$$(D_y)_{s5} = 2 B d$$

$$(D_x)_{s6} = 0 - (\frac{1}{2} B d) = - \frac{1}{2} B d$$

$$(D_y)_{s6} = 2 B d - (B d) = B d$$

For applying vertical shear force, F_y

$$q_1 = \frac{F_y}{I_x} (D_x)_{s1} = 0$$

$$q_2 = -\frac{1}{3} \frac{F_y}{d}$$

$$q_3 = 0$$

$$q_4 = \frac{1}{3} \frac{F_y}{d}$$

$$q_5 = 0$$

$$q_6 = -\frac{1}{3} \frac{F_y}{d}$$

Therefore, about shear centre,

$$\left(\frac{d\theta}{dz} \right)_{\text{cell 1}} = \frac{1}{2 \cdot d \cdot G} \frac{1}{t} (q_{01} \cdot 4d + q_1 \cdot d + q_2 \cdot d + q_3 \cdot d + q_4 \cdot d) = 0$$

$$\therefore 4dq_{01} - \frac{1}{3}F_y + \frac{1}{3}F_y = 0 \quad \therefore q_{01} = 0$$

Take moment at 1,

$$\begin{aligned} \therefore F_y \cdot X_E &= \oint p \cdot q \, ds = 2Aq_{01} + d \cdot q_2 \cdot d + d \cdot q_4 \cdot d - 2d \cdot q_6 \cdot d \\ &= 0 + 0 + \frac{1}{3}F_y d + \frac{2}{3}F_y d = F_y d \end{aligned}$$

$$\therefore X_E = d \text{ to the right of 1}$$

For applying horizontal shear force, F_x

$$q_1 = \frac{F_x}{I_y} (D_y)_{s1} = 0$$

$$q_2 = \frac{1}{4} \frac{F_x}{d}$$

$$q_3 = \frac{1}{2} \frac{F_x}{d}$$

$$q_4 = \frac{1}{2} \frac{F_x}{d}$$

$$q_5 = \frac{1}{2} \frac{F_x}{d}$$

$$q_6 = \frac{1}{4} \frac{F_x}{d}$$

Therefore, about shear centre,

$$\left(\frac{d\theta}{dz}\right)_{\text{cell 1}} = \frac{1}{2 \cdot d \cdot G} \frac{1}{t} (q_{01} \cdot 4d + q_1 \cdot d + q_2 \cdot d + q_3 \cdot d + q_4 \cdot d) = 0$$

$$\therefore 4dq_{01} + \frac{5}{4}F_x = 0 \quad \therefore q_{01} = -\frac{5}{16} \frac{F_x}{d}$$

Take moment at 1,

$$\begin{aligned} \therefore \underbrace{F_x \cdot Y_E}_{-F_x \cdot Y_E} &= \oint p \cdot q \, ds = 2Aq_{01} + d \cdot q_3 \cdot d + d \cdot q_4 \cdot d - 2d \cdot q_6 \cdot d \\ &= -\frac{5}{8}F_x d + \frac{1}{2}F_x d + \frac{1}{2}F_x d - \frac{1}{2}F_x d = -\frac{1}{8}F_x d \end{aligned}$$

$$\therefore Y_E = \underbrace{-\frac{1}{8}d}_{\frac{1}{8}d} \text{ above 1}$$